



Summer Internship 2025

Machine Learning Internship – Week 4 Tasks

Week 4 Task: Final Project Submission — Career Guidance Chatbot using Machine learning

Objective:

Build and submit a **Career Guidance Chatbot** that can understand user interests (via natural language input) and suggest relevant career fields using **Machine Learning-based career role classification** and a **Streamlit interface**.

Expected Deliverables:

1. A working chatbot with a **Streamlit frontend**
2. A trained **intent classification ML model**
3. Proper code structure and documentation
4. A short **video demo (recommended)**

Dataset Link (Use This for Training):

🔗 [Career Guidance Dataset \(CSV\)](#)

Steps to Follow

1. Understand the Dataset

Contains 1,620+ rows covering **54 different career roles**, with these columns:

- Role: Career role (e.g., “Data Scientist,” “Product Manager”)
- Question: User question about that role
- Answer: Informative response to guide career discussions

Explore the data using pandas. Clean or expand if needed.

2. Preprocess the Text

- Lowercase, remove punctuation, optional lemmatization
- Convert text into vectors using **TF-IDF**



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3. Train the career role Classifier

- Use a simple model like:
 - Logistic Regression
 - Naive Bayes
 - SVM
- Evaluate using **accuracy, F1-score**
- Save model and vectorizer using joblib or pickle

4. Build the Chatbot with Streamlit

- Create a web interface to:
 - Take user input
 - Predict intent using your model
 - Show a **career suggestion** based on predicted intent

Submission Instructions:

- Add comments, organize code into files

Project Folder Structure:

```
career_chatbot_project/  
├── app.py           # Streamlit frontend  
├── train_model.py   # Model training script  
├── career_guidance_dataset.csv  
├── intent_model.pkl # Saved model  
└── vectorizer.pkl   # Saved TF-IDF vectorizer
```

Send your Tasks via Gmail (nextgenlearners.official@gmail.com)